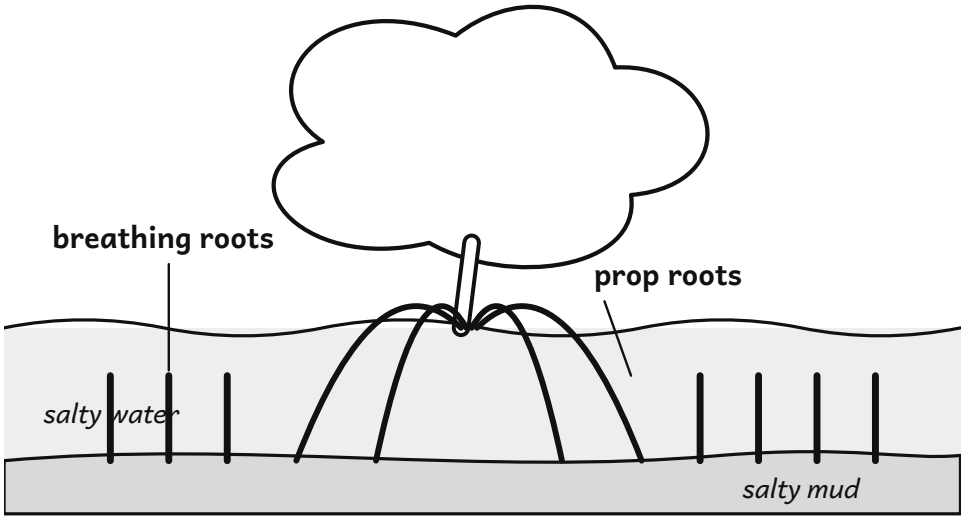
LIVING THINGS

Plants of Bangladesh: Diversity and Adaptation

Flowering and non-flowering plants, mangroves and floating plants, and how natural changes affect plants



In this pack

Learn it: Plants of Bangladesh and how they survive ★☆☆
Fact check: Plant groups and adaptations ★☆☆
Think it through: Why plants look the way they do ★★☆
Investigate: Does a waxy coat stop leaves drying out? ★★☆
Real-world case study: Water hyacinth: a problem or a gift? ★★★
Review and challenge: Show what you know ★★★

This pack belongs to: _____ **Class:** _____

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Plants of Bangladesh and how they survive

★☆☆ Grade 4

Name: Class: Date:

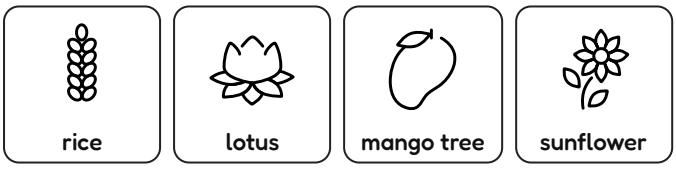
Learning goals

- Tell the difference between flowering and non-flowering plants.
- Describe adaptations of plants in different habitats of Bangladesh.
- Explain how natural changes can affect plants.

From the giant banyan tree in the village to the tiny duckweed floating on a pond, Bangladesh is green with plants. Plants give us food, wood, medicine, cloth and the oxygen we breathe. Every animal depends on them. Let's find out how many kinds there are, and how each one is suited to its home.

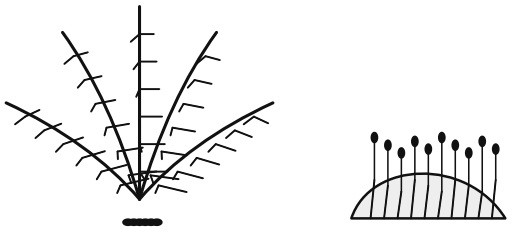
Flowering plants

Most plants you know are **flowering plants**. They grow flowers, which make **seeds**, often inside a **fruit**. Rice, mango trees, lotus, sunflowers, grass and the shapla (water lily, our national flower) are all flowering plants. Even grass and rice have flowers, but they are small and green.



Non-flowering plants

Some plants never make flowers. **Ferns** and **mosses** grow from tiny **spores** instead of seeds. Mosses grow in damp, shady places like the walls of old buildings. **Conifers**, such as pine trees, make seeds inside **cones**, not flowers.



Ferns and mosses make spores.

Did you know?

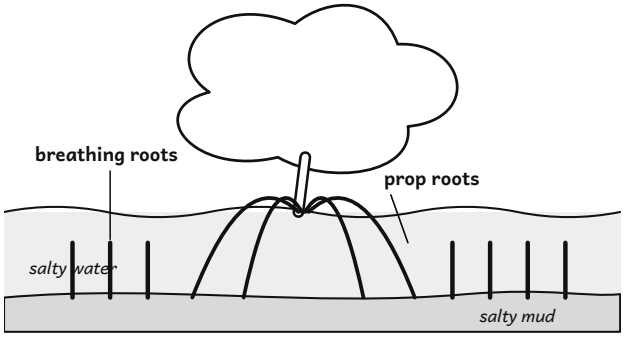
Mushrooms are not plants! They belong to a different group called **fungi**. They cannot make their own food from sunlight.

Adapted to their homes

An ***adaptation*** is a feature that helps a plant survive where it lives. Bangladesh has many different habitats, and the plants in each one are adapted in different ways.

Mangroves: living in salty mud

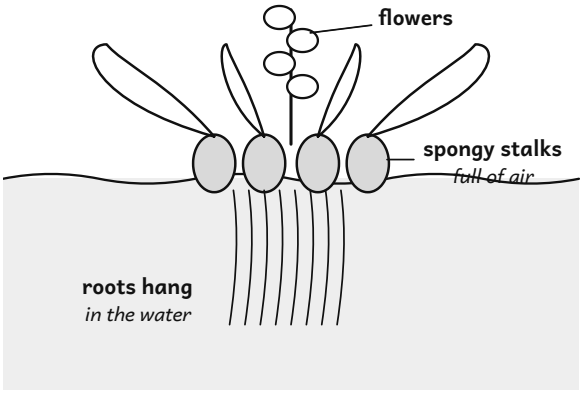
In the **Sundarbans**, the tide covers the ground with salty water twice a day. The mud has very little air. Mangrove trees such as the **sundari** (which gives the forest its name) have **breathing roots** that poke up out of the mud like pencils to take in air. Some have **prop roots** that hold the tree steady in soft mud. Their leaves can get rid of extra salt.



A mangrove tree.

Floating plants

Water hyacinth floats because its leaf stalks are swollen and spongy, full of air, like a float. Its roots hang down to take in minerals from the water. The **lotus** has a waxy leaf that water rolls off, and long hollow stems that carry air down to its roots in the mud.



Water hyacinth.

Dry and windy places

A **cactus** stores water in its thick stem and has **spines** instead of leaves, so it loses very little water. Coconut and palm trees on the coast have bendy trunks and long, split leaves, so storms do not snap them.

Natural changes

Floods can drown plants or bring rich new soil. **Droughts** dry them out. **Cyclones** blow trees down, and salty sea water pushed inland by storm surges can kill crops. Some plants survive; others die and new kinds may take their place. Over time the mix of plants in a place can change.

Bangladesh connection

After Cyclone Sidr in 2007, about a quarter of the Sundarbans was badly damaged, but the mangroves grew back over the following years. The Sundarbans also protects villages behind it by slowing the wind and waves.

Try this at home

Look for moss on a damp wall or tree trunk near your home. Use a hand lens to find the tiny capsules on stalks that hold the spores.

Key words

flowering plant	a plant that makes flowers and seeds
non-flowering plant	a plant with no flowers, such as a fern or moss
spore	a tiny cell that can grow into a new fern or moss
cone	the part of a pine tree that holds its seeds
mangrove	a tree that grows in salty mud on the coast
breathing roots	roots that grow up out of the mud to take in air
adaptation	a feature that helps a living thing survive
fungi	mushrooms and moulds: not plants

FACT CHECK · A2 · PLANT DIVERSITY

Plant groups and adaptations

★☆☆ Grade 4

Name: Class: Date:

Learning goals

- Sort plants into groups.
- Match adaptations to habitats.

1 Sort these living things. [8]

SORT THESE

mushroom fern mango tree rice mould on bread lotus pine tree moss

Flowering plant	Non-flowering plant	Not a plant

2 Match the plant to its adaptation. [5]

Draw a line to match each pair, or write the letter.

mangrove ●	● **A** thick stem stores water; spines instead of leaves
water hyacinth ●	● **B** waxy leaves that water rolls off
cactus ●	● **C** bendy trunk that does not snap in storms
lotus ●	● **D** spongy stalks full of air to float
coconut palm ●	● **E** breathing roots that stick up out of salty mud

3 Which of these living things is not a plant? [1]

A 

B 

C 

D 

◯ A ◯ B

◯ C ◯ D

4 **True or false?** [4]

Statement	True	False	If false, correct it
1. Ferns grow from spores.			
2. Rice plants do not have flowers.			
3. Mangrove trees live in salty water.			
4. A cyclone can bring salty water onto farmland.			

Why plants look the way they do

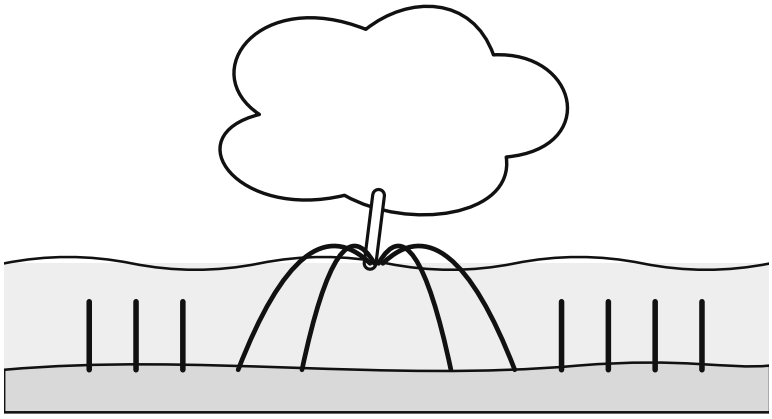
★★☆
Grade 4

Name: Class: Date:

Learning goals

- Explain adaptations.
- Predict the effects of change.

1 **Look at the mangrove.** [4]



a) Why does the mangrove need roots that grow **up** out of the mud?

b) How do prop roots help in a cyclone?

2 **Tisha says, "A cactus would grow well in the Sundarbans." Do you agree? Explain.** [2]

3 **A farmer's field near the coast is flooded by salty sea water after a storm. Predict what will happen to the rice and explain.** [2]

.....

.....

.....

4 **Design a plant that could live on a sunny, windy sand bank in a river (a char). Draw it and label two adaptations.** [3]

My plant

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Does a waxy coat stop leaves drying out?

★★☆ Grade 4

Name: Class: Date:

Learning goals

- Carry out a fair test.
- Measure mass.
- Explain results using adaptation.

Many leaves have a **waxy** coat. Class 4A made model leaves from wet paper towels. They rubbed petroleum jelly (like wax) on some and left others plain. They weighed them, left them in the sun, and weighed them again after 3 hours.

1 **Complete the plan.** [4]

Question	
Change	
Measure	
Keep the same	

2 **Here are the results.** [5]

Model leaf	Mass at start (g)	Mass after 3 hours (g)	Water lost (g)
plain 1	20	8	
plain 2	20	9	
waxy 1	20	17	
waxy 2	20	18	

a) Work out the water lost for each model leaf.

b) Which leaves lost less water?

.....

3 **Write a conclusion.** [2]

.....

.....

4 **Why did they test two of each kind of model leaf?** [1]

REAL-WORLD CASE STUDY · A2 · PLANT DIVERSITY

Water hyacinth: a problem or a gift?

★★★ Grade 4

Name: Class: Date:

Learning goals

- Weigh up good and bad effects.
- Link science to jobs and design.
- Give a reasoned opinion.

Science · Economics · Design · Environment

Water hyacinth (*kochuripana*) came from South America more than a hundred years ago. It grows very fast in warm, still water. It can double in size in about two weeks, covering ponds, canals and rivers like a green carpet.

This causes problems. Boats get stuck, fish have less light and air, and mosquitoes can breed in the still water underneath. But people in Bangladesh have found clever uses for it. Farmers pile it up to make **floating gardens** (*dhap*) to grow vegetables during floods. It is used as **compost** and cattle food, and craft workers weave the dried stems into baskets, mats and furniture that are sold around the world.

Based on information about water hyacinth and floating gardens in Bangladesh.

1 **Sort the effects.** [6]

SORT THESE

baskets and mats **floating gardens** **blocks boats** **mosquitoes breed** **compost for fields** **less light for fish**

Good for people	Bad for people or nature

2 **Which adaptation helps water hyacinth to cover a pond so fast?** [2]

3 **Maths. A patch of water hyacinth covers 1 square metre. It doubles every 2 weeks. How much will it cover after 8 weeks?** [2]

Weeks	0	2	4	6	8
Area (square metres)	1				

4 **Your opinion: Is water hyacinth a problem or a gift? Give reasons for both sides, then decide.** [3]

Show what you know

★★★
Grade 4

Name: Class: Date:

Learning goals

- Use key vocabulary.
- Answer test-style questions.

1 Find the key words.

F	L	W	F	H	A	D	A	P	T	R
G	L	F	E	R	N	A	A	M	T	M
F	J	M	O	R	W	Y	J	O	E	I
L	R	A	E	Y	S	R	O	O	T	S
O	D	N	C	B	C	F	A	I	A	I
W	S	G	O	E	V	E	T	K	V	C
E	A	R	R	O	L	S	E	E	D	A
R	D	O	T	F	H	J	B	K	S	C
M	F	V	I	N	A	M	O	S	S	T
H	F	E	K	H	I	Y	M	H	W	U
M	H	U	L	S	P	O	R	E	E	S

adapt cactus fern flower mangrove moss roots seed spore

2 Which plant makes spores instead of seeds? [1]

- ◯ rice
- ◯ fern
- ◯ mango tree
- ◯ lotus

3 Why does water hyacinth float? [1]

- ◯ its roots are heavy
- ◯ its stalks are full of air
- ◯ it has no leaves
- ◯ it is a fungus

4 Draw and label a plant from a dry place and a plant from a wet place. Show one adaptation on each. [4]

dry place

wet place

5 **Challenge. The sea level is rising and more salty water is reaching farms in the south of Bangladesh. How could scientists help farmers?** [2]

.....

.....

.....

How am I doing? Colour or tick a circle.

I can...	Not yet	Nearly	Got it!
tell flowering and non-flowering plants apart	○	○	○
explain that mushrooms are fungi, not plants	○	○	○
describe how mangroves and floating plants are adapted	○	○	○
investigate how a waxy coat saves water	○	○	○
explain how floods, droughts and cyclones change plant life	○	○	○
weigh up the good and bad effects of water hyacinth	○	○	○